TOSIN KOLAJO GBADEGESIN

Ph.D. in Economics
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RESEARCH INTERESTS

Energy Economics; Environmental Economics; Urban and Spatial Economics; Labor Economics; Causal Inference and Program Evaluation; Applied Machine Learning; Development Economics; Climate Policy; Regulatory Economics; Gender and Inclusion Economics.

EDUCATION

Ph.D. in Economics *2026*
Howard University, Washington, DC

- Fields: Energy and Environmental Economics; Urban and Regional Economics; Labor Economics; Causal Inference; Industrial Organization; Applied Econometrics
- Dissertation: "Climate Change, Energy Demand, and Policy Impacts: Analyzing Residential and Building Energy Dynamics in the U.S."
- Job Market Paper (Chapter 4): "The Building Energy Performance Standard Program: Impacts on Buildings' Energy Efficiency and GHG Emissions in Washington, DC," Applied Energy. [Link], [Replication Code]
- Advisor: Haydar Kurban, Ph.D. | Frederick Douglass Doctoral Scholar (2023-24) | World Bank Africa Doctoral Fellow

M.Sc. in Economic Development *2018*
University of Ibadan, Ibadan, Nigeria

B.Sc. in Microbiology (First Class Honours) *2013*
Obafemi Awolowo University, Ile-Ife, Nigeria

ACADEMIC AND PROFESSIONAL POSITIONS

Economist *January 2023 – Present*
World Bank Group (International Finance Corporation / PLANET), Washington, DC

- Lead economist for the Private Sector Gender and Economic Inclusion Data Platform, designing statistical measurement frameworks, automated ETL pipelines, and BI dashboards processing 2-4 TB of multi-source data.
- Designed and deployed econometric early warning systems (structural VAR and error-correction models) for food insecurity detection across fragile and conflict-affected states; analysis operationalized in the World Bank's Early Response Financing mechanism.
- Led causal impact evaluation of the G20 Compact with Africa (12 economies) using DiD, PSM, and Causal Forests; presented findings at the Germany-Africa Head of State Summit and World Bank Annual Meetings.
- Conducted applied market research on electricity sector structure and extractives performance; refined IFC's AIMM (Anticipated Impact Measurement and Monitoring) framework.
- Applied ensemble ML models (LightGBM, XGBoost, CatBoost) and Double ML on 500+ GB of IFC lending data for portfolio risk analysis.

Research Assistant | Teaching Assistant *2024 – 2025*
Department of Economics, Howard University, Washington, DC

- Research: Processed and analyzed 100M+ UN Comtrade international trade records using Python and SQL for a faculty-led research project.
- Teaching: Teaching Assistant, ECON 2101: Economic Statistics (undergraduate). Responsibilities included data-lab instruction, problem sets, and applied econometrics tutoring.

Energy Efficiency Analyst *June 2022 – December 2022*

DC Sustainable Energy Utility (DCSEU) and DC Department of Energy and Environment (DOEE), Washington, DC

- Conducted regulatory benefit-cost and causal policy analysis of DC's Building Energy Performance Standards (BEPS) across 100+ GB of commercial building panel data; produced regulatory guidance documents and stakeholder policy briefs.
- Designed automated analytics pipelines and dashboards for regulatory compliance reporting on decarbonization programs and the Solar for All initiative.

Research Analyst *September 2022 – December 2022*

Edelman Global Advisory, Washington, DC

- Delivered economic and regulatory analysis on trade and climate policy across six global regions; produced structured research reports for corporate and government clients.

Bridge Policy Research Fellow *August 2020 – August 2021*

Nigerian Economic Summit Group, Abuja, Nigeria

- Analyzed COVID-19 socioeconomic household impact survey data and reviewed national development plans for water and sanitation reform.
- Co-led establishment of the UN CEO Water Mandate in Nigeria; contributed technical analysis to medium-term national development plans and Agenda 2050.
- Contributed to national technical committees integrating gender and digital technology themes into development planning.

Development Finance Specialist *January 2020 – September 2020*

Jerusalem Institute for Policy Research | Gigawatt Global Inc., Israel

- Assessed pre-feasibility and feasibility of grid-connected solar infrastructure projects: solar yield analysis, transmission infrastructure evaluation, and ESG impact assessment.
- Built financial models evaluating returns, risks, and capital requirements; engaged development finance institutions and diplomatic stakeholders to mobilize project capital.

Data Analyst *September 2019 – July 2021*

International Institute for Local Development, Abuja, Nigeria

- Applied ML models (Random Forest, AdaBoost, KNN, SVM, CNN) to evaluate World Bank-funded rural access and agricultural development projects.
- Produced data visualization and analytical reports in R and Tableau for development decision-makers.

Research Fellow *July 2018 – December 2018*

Africa Water Association and USAID West Africa, Nigeria

- Researched urban water supply challenges; designed survey data collection pipelines (Kobo Collect); analyzed data in SPSS; built Tableau dashboards; presented findings at an international sustainable development conference.

Research Fellow *January 2018 – June 2018*

International Fund for Agricultural Development (IFAD), Arusha, Tanzania

- Investigated factors affecting smallholder farmers' market access using survey econometrics; published findings in peer-reviewed journals.

Team Lead, Projects and Programs *June 2017 – December 2020*

Clean Edge Exclusive Solution, Nigeria

- Directed project teams deploying mini-grid solar and energy efficiency solutions across Nigeria; convened the first Annual Youth Summit on Environment.
- Led national assessment of Nigeria's plastic circularity landscape in partnership with environmental organizations.

Research Analyst *June 2013 – June 2016*

Crown FM Broadcast Group and Independent Media, Nigeria

- Conducted audience and market research to guide multi-platform content strategy; analyzed digital performance metrics (SEO, engagement) to optimize audience reach.

PUBLICATIONS

Journal Articles

1. "The Building Energy Performance Standard Program: Impacts on Buildings' Energy Efficiency and GHG Emissions in Washington, DC". *Applied Energy*. [Link], [Replication Code]
2. "Beyond Affordability: A Spatial Econometric Analysis of Renter Cost Burdens and Social Determinants in the United States". *Economic Development Quarterly*. **[Accepted]**
3. "Defense Expenditure and National Well-Being: Evidence from the G7 Countries". *Defence and Peace Economics*, 2024, 36(7), 1030-1041. [Link]
4. "Evaluating the Economic Impacts of the G20 Compact Initiative: Evidence from Causal Inference Using Advanced Machine Learning Techniques" (with N. D. Yameogo). *Journal of Sustainable Development*, 2024, 17(5), 56-76. [Link]
5. "Climate Change and Energy Dynamics: Evidence from BRICS and G7 Countries". *Energy and Environment*, 2024, 0(0). [Link]
6. "Water Availability Challenges in Low-Income Areas of Agbowo Community, Ibadan, Nigeria" (with O. Olayide). *International Journal of Circular Economy and Waste Management*, 2021, 1(1), 81-96. [Link]
7. "Effectiveness of Collective Action in Reducing Transaction Cost for Smallholder Paddy Farmers in Tanzania" (with L. Popoola). *African Journal of Economic and Sustainable Development*, 2020, 7(4), 391-404. [Link]
8. "Transaction Cost Factors Affecting the Economic Life of Smallholder Paddy Farmers in Tanzania" (with L. Popoola and L. Muhoni), *Africa Journal of Sustainable Development*, 2019, 8(1). [Link]

Policy Research Working Papers

9. "Climate Shocks and Their Effects on Food Security, Prices, and Agricultural Wages in Afghanistan" (with B. P. J. Andree and A. Braimoh). *Policy Research Working Paper, World Bank Group (PLANET)*, 2024. [Link]

Book Chapters

10. "A Review on Impact of Changing Climate on Sustainable Food Consumption". *Global Food Politics and Approaches to Sustainable Consumption*, 2020, pp. 54-76. [Link]
11. "Climate Change Implication on Airborne Infections: Roadmap for Nigeria's Health Sector Development," in *Handbook of Climate Change Resilience*, 2018, 1–10. [Link]

Institutional Reports

12. Closing the Gap: A Data Outlook on Women's Economic Opportunities. *International Finance Corporation*, 2025. [Link]
13. Repurposing Power Markets: The Path to Sustainable and Affordable Energy for All. *International Finance Corporation*, 2024. [Link]

14. Compact Monitoring Report: Africa Advisory Group Meeting. *World Bank Group / G20 Compact with Africa*, 2023. [Link]

Blog

15. "From data gaps to action: measuring women's economic opportunities better" (with N. Datta and M. Nettikumara), *World Bank Blogs*, 2026. [Link]
16. "Better Data for Better Outcomes: New Tools and Analysis to Break Afghanistan's Food Security Crisis" (with B. P. J. Andree and A. Braimoh), *World Bank Blogs*, 2025. [Link]

WORKING PAPERS AND WORKS IN PROGRESS

Under Review

17. "Heterogeneous Effects of Cooling Demand on Household Electricity Consumption using Causal Machine Learning Models". *Applied Energy*.
18. "Impact of Climate-Driven Cooling Demand on U.S. Household Electricity Consumption and Expenditure". *Energy and Environment*.

Works in Progress

- "Labor Market Impacts of AI Adoption: A Causal Evaluation of Employment and Wage Effects Across Occupational Groups."

CONFERENCE PRESENTATIONS AND INVITED TALKS

Conference Presentations

- "Building Energy Performance Standards and GHG Emissions." *Empirical Methods in Energy Economics (EMEE), World Bank Group*, Washington, DC. 2025.
- "Collective Action and Smallholder Market Access in Tanzania." *Development Studies Association Annual Conference, Trinity College Dublin*, Dublin, Ireland. 2019.

TEACHING EXPERIENCE

Teaching Assistant -- Howard University

ECON 2101: Economic Statistics (Undergraduate) *Spring 2025*

ECON 2101: Economic Statistics (Undergraduate) *Fall 2024*

Responsibilities: Data-lab instruction using R and Python; problem set grading; applied econometrics tutoring; supplemental lectures on statistical inference and regression analysis.

Guest Lectures and Workshops

"Causal Inference for Policy Analysis," Howard University Graduate Seminar *2024*

"Applied Machine Learning in Economics," World Bank Group Internal Workshop *2023*

GRANTS, FELLOWSHIPS, AND AWARDS

World Bank Group Africa Doctoral Fellowship. *World Bank Group -- 1 of 32 fellows selected from 3,000+ applications worldwide* 2023

Frederick Douglass Doctoral Scholarship. *Howard University Graduate School of Arts and Sciences* 2023-2024

Blum Global Fellowship in Development Finance. *Jerusalem Institute for Policy Research and Hebrew University Business School, Israel* 2020

Early Career Conference Grant. *Association of Commonwealth Universities (ACU), London, United Kingdom* 2019

Young African Leaders Initiative (YALI) Leadership Award in Energy and Environmental Sustainability. *YALI Network, Nigeria* 2018

Young IFAD Scholar. *International Fund for Agricultural Development* 2018

PROFESSIONAL TRAINING AND CERTIFICATIONS

Artificial Intelligence with Python. *Harvard University Extension School,* 2023.

Post Graduate Program in Data Science. *Career Era and Birchwood University,* 2022.

Data Analytics Professional Certificate. *Google,* 2021.

Machine Learning Engineering. *Amazon Web Services (AWS),* 2021.

Blum Global Fellowship in Development Finance Practice. *Hebrew University Business School and Jerusalem Institute for Policy Research,* 2020.

PROFESSIONAL AFFILIATIONS AND SERVICE

Professional Memberships

- American Economic Association (AEA)
- National Association for Business Economics (NABE)

Reviewer

- Energy and Environment
- Applied Energy
- Defence and Peace Economics

Institutional Service

- Technical contributor, Nigeria Agenda 2050 national development framework, 2020-2021
- Co-lead, UN CEO Water Mandate Nigeria chapter establishment, 2020-2021
- Convener, First Annual Youth Summit on Environment, Nigeria, 2018

TECHNICAL SKILLS

Econometrics and Causal Inference: Panel and Time-Series Models; Instrumental Variables and GMM; Spatial Econometrics; Difference-in-Differences; Regression Discontinuity Design; Synthetic Control; Propensity Score Matching; Quantile Regression; Causal Forests; Double Machine Learning; Conditional Average Treatment Effects (CATE); Conditional Distributional Treatment Effects (CDTE); Heterogeneous Treatment Effects.

Machine Learning and Data Science: XGBoost, LightGBM, CatBoost, Random Forest, AdaBoost, Super Learner, SVM, CNN; NLP and LLM; Feature Engineering and Model Selection; A/B Testing and Experimental Design; Large-Scale Data Processing (500GB+); Automated Pipeline Design.

Programming Languages and Software: Python (pandas, NumPy, statsmodels, scikit-learn, EconML, Matplotlib, Seaborn); R (tidyverse, ggplot2, fixest, rdrobust); Stata; SQL; MATLAB; SPSS; PVSYST; Google Earth Engine; Power BI; Tableau; ArcGIS; LaTeX; GitHub; Microsoft Azure; Amazon Web Services.